FLEET TEAM DIGEST

737MAX-FTD-32-17012

Issue Title : Nose Landing Gear Wheel Bearing Corrosion Damage

Airplane Model	ATA	Minor Model(s)
737MAX	3240-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/02/2017	05/02/2017	04/21/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX airplanes.

Description

A 737NG-BBJ operator reported two cases where they heard loud noises coming from a nose landing gear wheel and tire assembly during taxi. In both cases, the cause of the noise was traced to a damaged P/N L507949 nose wheel inner bearing. The bearings were nearly seized due to heavy corrosion. The grease was found to be heavily contaminated with water and rust. The operator advised that airplane had accumulated only 114 flight cycles since airplane delivery two years earlier. They also reported that they had not pressure washed the airplane.

Another 737NG operator reported having difficulty removing the nose wheels because the bearings were frozen (rusted) onto the axle. The nose wheels had between 673 and 766 landings since the last installation.

Hours/Cycles

234 Flight Hours/ 114 Flight Cycles

Background

Contamination of the bearing with water may be the combined result of factors such as overzealous airplane pressure washing, inconsistent use of wheel covers during washing, damaged bearing grease seals, and/or extended periods of airplane inactivity (condensation).

It is our understanding that bearings can collect moisture simply from condensation from the surrounding air. This situation is more likely to occur in areas where the climate tends to have high relative humidity.

Under normal airplane operating circumstances, it appears that if the wheels are turned regularly, any moisture will not sit long enough in one spot to cause significant corrosion damage. Moisture that does condense out of the atmosphere is apparently dissipated by the combined effect of the rotating action of the bearing elements and the resulting heat generated.

To avoid the collection of water in the wheel bearings, it is our understanding that some 737-BBJ operators jack the axles every three to four days to rotate the tires to ensure that bearing elements are exercised. This practice also ensures that the tires do not develop flat spots from resting on same portion of the tire all the time. Also, it may be necessary to remove the wheels

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periodically and inspect/regrease the bearings.

Examination of the bearings removed from the airplanes where the wheels were frozen onto the axle revealed that the corrosion was on the inside diameter of the bearing, where the bearing contacts the axle journal. This was likely the result of inadequate lubrication in the area, combined with the fact that bearing was installed on the airplane for an unusually long time.

Status

Closed

Interim Action

See "Operator Action" below

Final Action

See "Operator Action" below

Operator Action

Operators are encouraged to periodically remove and examine the wheels on low utilization airplanes to look for evidence of rust, water, or other foreign material. We do not have specific recommendations for how often this task should be performed since it is likely to be highly dependent on the climate, airplane utilization, and whether the airplane is stored inside or outside. Each operator will have to select an appropriate interval based on findings during these inspections.

We also recommend that all operators consider changing the type of grease used on the wheel bearings. Several operators have had good experience with Mobil SHC 100 aviation grease which is reported to help prevent moisture from reaching the bearing surfaces. The use of SHC 100 is approved by Goodrich per service letter 1777.

For operators who obtain high numbers of landings on their nose gear tires, we recommend that they pack the nose wheel bearings with generous amounts of grease and lubricate the axle journals with generous amounts of grease where the inside diameter of the bearing contacts the axle. This will provide additional corrosion protection and help keep the bearing lubricated longer.

For added protection against water contamination of wheel bearings, we suggest that operators ensure that the wheels and brakes are protected (covered) during airplane washing (see AMM 12-40-00/201) or at a minimum, the wheels are not directly sprayed with water when washing.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Letter (SL)	BFG SL 1777		05/06/1998
Aircraft Maintenance Manual (AMM)	10-12-00/201		
Aircraft Maintenance Manual (AMM)	12-40-00/201		

Related Categories

- In Service Occurrences : null

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Attachments:

a) Inner Race Diameter Corrosion

b) Corroded Nose Wheel Bearing

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